

W12,L02: Resampling and Splines

Lecture Notes

1 Resampling

1.1 Introduction

Want to do inference: estimation/testing

Data: $X_1, \dots, X_n$ iid $F(\theta)$, where $X \sim F$ iid. We have $\int xf(x)dx = \mu_x$.

Problem: F is unknown. No parametric model.

Challenge: We may not have non-parametric methods available.

We have CI for median using non-parametric methods. We might be interested in other parameters like mean, [You can find MoM estimate as $\bar{X}$. But what is its distribution? How to obtain confidence intervals or test $\mu = \mu_0$?].

Skewness, variance, quantile? (Non-parametric)

Even if we have parametric/non-parametric methods we can still use resampling methods and in some cases get better performance.

Q: How to obtain CI for median using non-parametric method?

1.2 Jackknife Method

1.2.1 Definition

$X_1, \dots, X_n \sim F$. θ = skewness of F .

$\hat{\theta}_n$ = sample skewness.

Jackknife:

$\hat{\theta}_{(-1)}$ = sample skewness of $\{X_2, \dots, X_n\}$

$\hat{\theta}_{(-2)}$ = sample skewness of $\{X_1, X_3, \dots, X_n\}$

$\hat{\theta}_{(-i)}$ = remove X_i from sample left sample skewness of remaining $(n-1)$ data points.

Bias:

$$\bar{\theta}_{(o)} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}$$

1.2.2 Jackknife Estimator of θ

New estimator (Jackknife estimator) of θ is:

$$\tilde{\theta}_n = \hat{\theta}_n + (n-1)(\hat{\theta}_n - \hat{\theta}_{(o)}) = n\hat{\theta}_n - (n-1)\hat{\theta}_{(o)}$$

Claim: $\tilde{\theta}_n$ has lower bias than $\hat{\theta}_n$ (under assumptions)

1.2.3 Resampling Assumption

Resampling: we are taking many samples from the original sample.

Assumption:

$$E[\hat{\theta}_n] = \theta + \frac{a_1}{n} + \frac{a_2}{n^2} + o(n^{-2})$$

Often true if $\tilde{\theta}_n$ is MLE.

Proof of claim under the assumption:

$$E[\hat{\theta}_{(-i)}] = \theta + \frac{a_1}{n-1} + \frac{a_2}{(n-1)^2} + o(n^{-2})$$

$\hat{\theta}_{(-i)}$ is same as previous one but with $n-1$ independent observations.

$$\mathbb{E}[\bar{\theta}_{(o)}] = \frac{1}{n} \sum_{i=1}^n E[\hat{\theta}_{(-i)}] = \theta + \frac{a_1}{n-1} + \frac{a_2}{(n-1)^2} + o(n^{-2})$$

$$\begin{aligned} E[\tilde{\theta}_n] &= nE[\hat{\theta}_n] - (n-1)E[\hat{\theta}_{(-)}] \\ &= n\theta + a_1 + \frac{a_2}{n} + o(n^{-1}) - (n-1)\theta - a_1 - \frac{a_2}{n-1} + o(n^{-1}) \\ &= \theta - \frac{a_2}{n(n-1)} + o(n^{-1}) \end{aligned}$$

$$E[\hat{\theta}_n] = \theta + \frac{a_1}{n} + o(n^{-1}) = \theta + O(n^{-1})$$

$$E[\tilde{\theta}_n] = \theta - \frac{a_1}{n(n-1)} + o(n^{-1}) = \theta + o(n^{-1})$$

Jackknife method: Find $\hat{\theta}_n$ based on $X_1, \dots, X_n$

$\hat{\theta}_{(-i)}$ based on $X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n$

$$\tilde{\theta}_{(o)} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}$$

Jackknife estimator:

$$\tilde{\theta}_n = n\hat{\theta}_n - (n-1)\bar{\theta}_{(o)}$$

If

$$\mathbb{E}[\hat{\theta}_n] = \theta + \frac{a_1}{n} + \frac{a_2}{n^2} + o(n^{-2})$$

then bias of $\tilde{\theta}_n$ is less than bias of $\hat{\theta}_n$.

$$\bar{\theta}_{(o)} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)}$$

$$\hat{\theta}_{(-1)}, \dots, \hat{\theta}_{(-n)}$$

Claim:

$$\text{Var}(\hat{\theta}_n) \approx \frac{1}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n \left(s_{j(-i)} - \hat{\theta}_{(o)} \right)^2$$

(sample variance)

pseudo observations:

$$\tilde{s}_{ji} = \hat{\theta}_n + (n-1) \left(\hat{\theta}_n - \hat{\theta}_{(-i)} \right)$$

$$\frac{1}{n} \sum_{i=1}^n \tilde{s}_{ji} = \hat{\theta}_n + (n-1) \left(\hat{\theta}_n - \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(-i)} \right) = \tilde{\theta}_n$$

Jackknife: systematically remove 1 data point to get pseudo observations s_j

sample median s_j is bias corrected estimator

$\text{Var}(s_j)$ estimates the variance of $\hat{\theta}$.

Bootstrap: $X_1, \dots, X_n \sim F$ $\hat{\theta}_n$, θ parameter of interest

Randomly pick a single random sample of size n from the original sample with replacement

$$\{X_1^*, \dots, X_n^*\} = \hat{\theta}_1^*$$

for $n=5$

$$X_5, X_2, X_5, X_1, X_2$$

a bootstrap sample

Repeat: Obtain B bootstrap samples

$$\hat{\theta}_1^*, \dots, \hat{\theta}_B^* \rightarrow \text{bootstrap estimates of } \hat{\theta}$$

Ideally we want n large $\rightarrow$ lots of sample points from F (cdf)

Cannot do that

We draw samples from $\hat{F}_n$ the empirical cdf

$\hat{F}_n$ is an approximation to F

$\hat{\theta}_i^*$ are like multiple observations of $\hat{\theta}_n$ from $\hat{F}_n$

These give an approximation to the distribution of $\hat{\theta}_n$

gives us $\hat{\theta}_i^*$'s CI and cut-off for hypothesis tests 95%

Let $X_1, \dots, X_n \stackrel{iid}{\sim} F$, with parameter estimate $\hat{\theta}$.

$$\begin{aligned}\hat{\theta}_1^* &\leftarrow \{X_{11}^*, \dots, X_{1n}^*\} \quad \text{Simple random sample with replacement} \\ \hat{\theta}_2^* &\leftarrow \{X_{21}^*, \dots, X_{2n}^*\} \\ &\vdots \\ \hat{\theta}_B^*\end{aligned}$$

From these samples, we obtain:

- $\bar{\theta}^*$ is the bootstrap estimate of θ
- $sd(\hat{\theta}^*)$
- Histogram of $\hat{\theta}^*$'s (Quantiles, Cut-off for tests & CI)

1

Assume a form for F with parameter $\theta(F)$:

- Parametric estimation of θ may or may not be possible.
- Distribution of the parametric estimator may not be available.

Suppose F is parametrized by η . Estimate η parametrically / MoM / MLE as $\hat{\eta}$.

1. Generate $X_1^*, \dots, X_n^*$ from $F(\hat{\eta})$.
2. Find $\hat{\theta}_1^* \leftarrow \{X_{11}^*, \dots, X_{1n}^*\}$.
3. Repeat as in Bootstrap.

Difference: Instead of sampling from the empirical distribution $\hat{F}_n$, sample from the parametric distribution $F(\hat{\eta})$.

This gives the distribution of $\hat{\theta}^*$, cutoffs, etc.

2 Spline Regression

2.1 Non-parametric Regression

Model:

$$Y = f(x) + \varepsilon$$

We do not know anything about f .

Want: estimate f .

Available data: $(x_1, y_1), \dots, (x_n, y_n)$.

Kernel methods. Weighted average of y 's.

The idea of splines is that we want f to be smooth.

2.1.1 Cubic Splines

We fix $\xi_1, \dots, \xi_k$ (not necessarily related to $x_1 \cdots x_n$) as knots. $x_{(1)} < \cdots < x_{(n)}$. Do not want ξ_j to be too far from $[x_{(1)}, x_{(n)}]$.

We require that f satisfies:

- (i) f is a cubic polynomial between $(\xi_j, f(\xi_j))$ and $(\xi_{j+1}, f(\xi_{j+1}))$ has (not necessarily related to $x_1 \cdots x_n$) for $j=1, 2, \dots, k-1$.
- (ii) f is continuous at the knots
- (iii) f has continuous 1st and 2nd derivative at each knot

Instead of specified cubic, in general, splines can be of any specified degree.

Among all f 's satisfying above, choose the one that minimizes:

$$\sum_{i=1}^n (y_i - f(x_i))^2$$

Result: We can write f in terms of $k + 3$ basis functions:

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 h(x - \zeta_1) + \beta_5 h(x - \zeta_2) + \cdots + \beta_{k+3} h(x - \zeta_k)$$

$$h(u) = u^3 I(u > 0)$$

How to choose k and $\zeta_1, \dots, \zeta_k$:

- The ζ_i 's are chosen as quantiles of the x 's.
- k is chosen by cross-validation (CV).

Natural cubic spline:

A spline which is linear for $x < \zeta_{(1)}$ and $x > \zeta_{(k)}$, but cubic in between, is called a *natural cubic spline*.

2.2 Smoothing Splines

$$\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int_{x_{(1)}}^{x_{(n)}} [f''(x)]^2 dx$$

Provides a balance between the 2 goals of fit and smoothness. Called smoothness penalty/parameter.

Choose λ by cross validation.

Result: Solution of smoothing spline is the natural cubic spline.